



POWERED BY
CYBER SKYLINE

The National Cyber League
A Community Where Cybersecurity Is a Passion

Matthew Grote
mgrote1@my.wgu.edu

NCL Fall 2021 Team Game Scouting Report

Dear Matthew Grote (Team "Umbrella Corporation | WGU"),

Thank you for participating in the National Cyber League (NCL) 2021 Fall Season! Our goal is to prepare the next generation of cybersecurity professionals, and your participation is helping achieve that goal.

The NCL was founded in May 2011 to provide an ongoing virtual training ground for collegiate students to develop, practice, and validate their cybersecurity skills in preparation for further learning, industry certifications, and career readiness. The NCL scenario-based challenges were designed around performance-based exam objectives of CompTIA certifications and are aligned to the National Initiative for Cybersecurity Education (NICE) Cybersecurity Workforce Framework published by the National Institute of Standards and Technology (NIST).

As you look to a future career in cybersecurity, we hope you find this report to be valuable in both validating skills and identifying areas for improvement across the nine NCL skills categories. You can use this NCL Scouting Report to:

- Validate your skills to employers in any job application or professional portfolio;
- Show case your achievements and strengths by including the Score Card view of your performance as part of your résumé or simply sharing the validation link so that others may view the detailed version of this report.

The NCL 2021 Fall Season had 7,130 students/players and 491 faculty/coaches from more than 500 two- and four-year schools & 70 high schools across all 50 U.S. states registered to play. The Individual Game Capture the Flag (CTF) event took place from October 22 through October 24. The Team Game CTF event took place from November 5 through November 7. The games were conducted in real-time for students across the country.

NCL is powered by Cyber Skyline's cloud-based skills evaluation platform. Cyber Skyline hosted the scenario-driven cybersecurity challenges for players to compete and track their progress in real-time.



To validate this report, please access: cyberskyline.com/report/EKVTH8P45WVM

CompTIA. Based on the the performance detailed in this NCL Scouting Report, you have earned **16 hours** of Continuing Education Units (CEUs) as approved by CompTIA. You can learn more about the NCL - CompTIA alignment via nationalcyberleague.org/comptia.

Congratulations for your participation in the NCL 2021 Fall Team Game! We hope you will continue to develop your knowledge and skills and make meaningful contributions as part of the Information Security workforce!

David Zeichick
NCL Commissioner



POWERED BY
CYBER SKYLINE

NATIONAL CYBER LEAGUE SCORE CARD

NCL 2021 FALL TEAM GAME

YOUR TOP CATEGORIES

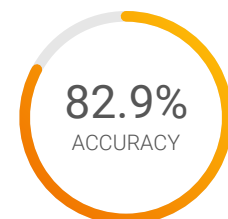


NATIONAL RANK

89TH PLACE
OUT OF 3910

PERCENTILE

98TH



Average: 62.4%

[cyberskyline.com/report](https://cyberskyline.com/report/EKVTH8P45WVM)
ID: EKVTH8P45WVM

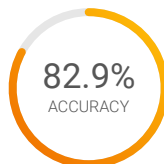


NCL Fall 2021 Team Game

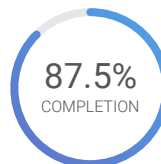
The NCL Team Game is designed for student players nationwide to compete in realtime in the categories listed below. The Team Game promotes camaraderie and evaluates the collective technical cybersecurity skills of the team members.

89TH PLACE
OUT OF 3910
NATIONAL RANK

2325 POINTS
OUT OF 3000
PERFORMANCE SCORE



Average: 62.4%



Average: 42.9%

98th National
Percentile

Average: 1052.4 Points

Cryptography

180 POINTS
OUT OF 300

100.0%
ACCURACY

COMPLETION: **78.6%**

Identify techniques used to encrypt or obfuscate messages and leverage tools to extract the plaintext.

Enumeration & Exploitation

240 POINTS
OUT OF 320

100.0%
ACCURACY

COMPLETION: **85.7%**

Identify actionable exploits and vulnerabilities and use them to bypass the security measures in code and compiled binaries.

Forensics

310 POINTS
OUT OF 310

77.8%
ACCURACY

COMPLETION: **100.0%**

Utilize the proper tools and techniques to analyze, process, recover, and/or investigate digital evidence in a computer-related incident.

Log Analysis

320 POINTS
OUT OF 320

84.2%
ACCURACY

COMPLETION: **100.0%**

Utilize the proper tools and techniques to establish a baseline for normal operation and identify malicious activities using log files from various services.

Network Traffic Analysis

285 POINTS
OUT OF 360

91.7%
ACCURACY

COMPLETION: **88.0%**

Identify malicious and benign network traffic to demonstrate an understanding of potential security breaches.

Open Source Intelligence

310 POINTS
OUT OF 310

64.7%
ACCURACY

COMPLETION: **100.0%**

Utilize publicly available information such as search engines, public repositories, social media, and more to gain in-depth knowledge on a topic or target.

Password Cracking

270 POINTS
OUT OF 350

95.2%
ACCURACY

COMPLETION: **83.3%**

Identify types of password hashes and apply various techniques to efficiently determine plaintext passwords.

Scanning & Reconnaissance

190 POINTS
OUT OF 310

73.3%
ACCURACY

COMPLETION: **73.3%**

Identify and use the proper tools to gain intelligence about a target including its services and potential vulnerabilities.

Web Application Exploitation

120 POINTS
OUT OF 320

100.0%
ACCURACY

COMPLETION: **50.0%**

Identify actionable exploits and vulnerabilities and use them to bypass the security measures in online services.

Note: Survey module (100 points) was excluded from this report.



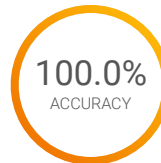


Cryptography Module

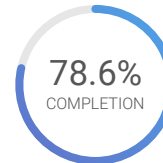
Identify techniques used to encrypt or obfuscate messages and leverage tools to extract the plaintext.

119 TH PLACE
OUT OF 3910
NATIONAL RANK

180 POINTS
OUT OF 300
PERFORMANCE SCORE



Average: 84.8%



Average: 58.5%

Security Control Assessor
Secure Software Assessor
Exploitation Analyst
Cyber Operator
Security Architect

97th National
Percentile

Average: 132.4 Points

Decoding 1 (Easy)

30 POINTS
OUT OF 30

100.0%
ACCURACY

COMPLETION: **100.0%**

Identify the type of base encoding used and decode the data

Decoding 2 (Easy)

20 POINTS
OUT OF 20

100.0%
ACCURACY

COMPLETION: **100.0%**

Identify the cipher scheme used and decrypt the data

Decoding 3 (Medium)

20 POINTS
OUT OF 20

100.0%
ACCURACY

COMPLETION: **100.0%**

Identify the cipher scheme used and decrypt the data

Emojis (Medium)

50 POINTS
OUT OF 50

100.0%
ACCURACY

COMPLETION: **100.0%**

Identify how emojis can be used to hide and store IP addresses

Mixtape (Medium)

60 POINTS
OUT OF 60

100.0%
ACCURACY

COMPLETION: **100.0%**

Identify the technique used to hide data in an audio file and extract it

PEM (Hard)

0 POINTS
OUT OF 120

0.0%
ACCURACY

COMPLETION: **0.0%**

Recover a redacted PEM key to its original version by exploiting the redundancies in PEM keys



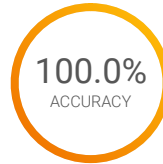


Enumeration & Exploitation Module

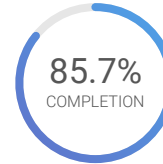
Identify actionable exploits and vulnerabilities and use them to bypass the security measures in code and compiled binaries.

111TH PLACE
OUT OF 3910
NATIONAL RANK

240 POINTS
OUT OF 320
PERFORMANCE SCORE



Average: 37.3%



Average: 41.5%

Cyber Operator
Target Developer
Exploitation Analyst
Software Developer
Systems Security Analyst

98th National
Percentile

Average: 108.4 Points

Fancy (Easy)

100 POINTS
OUT OF 100

100.0%
ACCURACY

COMPLETION: **100.0%**

Analyze a script source code & reverse engineer its functionality

Cell (Medium)

100 POINTS
OUT OF 100

100.0%
ACCURACY

COMPLETION: **100.0%**

Analyze a Haxe (Java runtime compatible) program & reverse engineer its functionality

Moblin (Hard)

40 POINTS
OUT OF 120

100.0%
ACCURACY

COMPLETION: **66.7%**

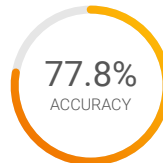
Decompile and analyze a C++ binary file for ARM & reverse engineer its functionality

Forensics Module

Utilize the proper tools and techniques to analyze, process, recover, and/or investigate digital evidence in a computer-related incident.

57TH PLACE
OUT OF 3910
NATIONAL RANK

310 POINTS
OUT OF 310
PERFORMANCE SCORE



Average: 46.2%



Average: 43.3%

Cyber Defense Forensics
Analyst
Cyber Crime Investigator
Cyber Defense Incident
Responder
Cyber Defense Analyst

99th National
Percentile

Average: 139.8 Points

File Recovery (Easy)

100 POINTS
OUT OF 100

100.0%
ACCURACY

COMPLETION: **100.0%**

Recover lost files from a NTFS filesystem

RAID (Medium)

100 POINTS
OUT OF 100

100.0%
ACCURACY

COMPLETION: **100.0%**

Recover data from a damaged RAID 5 disk array

Game Data (Hard)

110 POINTS
OUT OF 110

69.2%
ACCURACY

COMPLETION: **100.0%**

Analyze and carve the binary save data from a Game Boy videogame





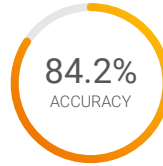
Log Analysis Module

Utilize the proper tools and techniques to establish a baseline for normal operation and identify malicious activities using log files from various services.

35TH PLACE
OUT OF 3910
NATIONAL RANK

320 POINTS
OUT OF 320
PERFORMANCE SCORE

Average: 144.3 Points



Average: 54.9%



Average: 55.9%

Cyber Defense Analyst
Systems Security Analyst
All-Source Analyst
Cyber Defense Forensics
Analyst
Data Analyst

100th National
Percentile

Backup (Easy)

100 POINTS
OUT OF 100

72.7%
ACCURACY

COMPLETION: **100.0%**

Analyze a SQL backup file to identify trends & locate sensitive information

Toasty (Medium)

100 POINTS
OUT OF 100

100.0%
ACCURACY

COMPLETION: **100.0%**

Parse the log file to recreate the sequence events & identify what happened

IOT Sensors (Hard)

120 POINTS
OUT OF 120

100.0%
ACCURACY

COMPLETION: **100.0%**

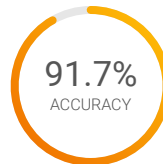
Network Traffic Analysis Module

Identify malicious and benign network traffic to demonstrate an understanding of potential security breaches.

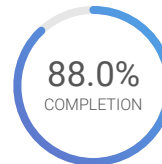
82ND PLACE
OUT OF 3910
NATIONAL RANK

285 POINTS
OUT OF 360
PERFORMANCE SCORE

Average: 173.7 Points



Average: 64.6%



Average: 52.2%

Cyber Defense Analyst
All-Source Analyst
Cyber Defense Incident
Responder
Target Network Analyst
Cyber Operator

98th National
Percentile

Scanning Activity (Easy)

80 POINTS
OUT OF 80

100.0%
ACCURACY

COMPLETION: **100.0%**

Analyze a capture of SMTP traffic to identify access IPs and user credentials

Cracking (Medium)

70 POINTS
OUT OF 70

100.0%
ACCURACY

COMPLETION: **100.0%**

Analyze and identify WiFi network metadata and crack the WiFi password

Jackbox (Medium)

100 POINTS
OUT OF 100

90.9%
ACCURACY

COMPLETION: **100.0%**

Analyze the HAR capture of websocket traffic of a video game

Remote Control (Hard)

35 POINTS
OUT OF 110

75.0%
ACCURACY

COMPLETION: **50.0%**

Identify the protocol used for sending IR over IP and perform custom dissection on the network data fields





Open Source Intelligence Module

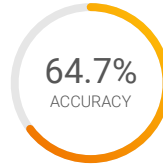
Utilize publicly available information such as search engines, public repositories, social media, and more to gain in-depth knowledge on a topic or target.

132 ND PLACE
OUT OF 3910
NATIONAL RANK

310 POINTS
OUT OF 310
PERFORMANCE SCORE

97th National
Percentile

Average: 212.7 Points



Average: 69.1%



Average: 77.1%

Systems Security Analyst
Target Developer
System Administrator
Research & Development
Specialist
Cyber Intel Planner

Rules of Conduct (Easy)

25 POINTS
OUT OF 25

100.0%
ACCURACY

COMPLETION: **100.0%**

Router Spec (Easy)

80 POINTS
OUT OF 80

100.0%
ACCURACY

COMPLETION: **100.0%**

Identify hardware specifications of a router device

Vessels Tracking (Easy)

70 POINTS
OUT OF 70

71.4%
ACCURACY

COMPLETION: **100.0%**

Identify & locate a naval vessel using public data sources

Shopping List (Medium)

45 POINTS
OUT OF 45

100.0%
ACCURACY

COMPLETION: **100.0%**

Find patterns in a set of item numbers to locate the target retailer

Satellite Tracking (Hard)

90 POINTS
OUT OF 90

28.6%
ACCURACY

COMPLETION: **100.0%**

Calculate or look up the orbit of a weather satellite



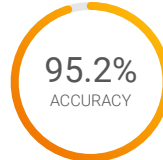


Password Cracking Module

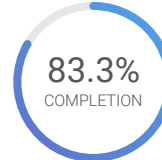
Identify types of password hashes and apply various techniques to efficiently determine plaintext passwords.

119TH PLACE
OUT OF 3910
NATIONAL RANK

270 POINTS
OUT OF 350
PERFORMANCE SCORE



Average: 88.2%



Average: 43.3%

Cyber Operator
Exploitation Analyst
Systems Security Analyst
Cyber Defense Incident
Responder
Cyber Crime Investigator

97th National
Percentile

Average: 128.5 Points

Cracking 1 (Easy)

Crack MD5 password hashes

30 POINTS
OUT OF 30

100.0%
ACCURACY

COMPLETION: **100.0%**

Cracking 2 (Easy)

Crack Windows NTLM password hashes

30 POINTS
OUT OF 30

100.0%
ACCURACY

COMPLETION: **100.0%**

Cracking 3 (Medium)

Identify patterns in the passwords and utilize a non-standard wordlist

15 POINTS
OUT OF 45

50.0%
ACCURACY

COMPLETION: **33.3%**

Cracking 4 (Hard)

Build a wordlist to crack passwords following a specific pattern

95 POINTS
OUT OF 95

100.0%
ACCURACY

COMPLETION: **100.0%**

Cracking 5 (Hard)

Build a wordlist to crack passwords not found in common wordlists

30 POINTS
OUT OF 80

100.0%
ACCURACY

COMPLETION: **60.0%**

PDF (Medium)

Crack the password of a PDF file

30 POINTS
OUT OF 30

100.0%
ACCURACY

COMPLETION: **100.0%**

Kali Linux (Hard)

Crack the new yescrypt password hashes included in Kali Linux

40 POINTS
OUT OF 40

100.0%
ACCURACY

COMPLETION: **100.0%**





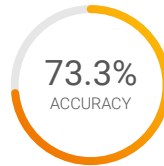
Scanning & Reconnaissance Module

Identify and use the proper tools to gain intelligence about a target including its services and potential vulnerabilities.

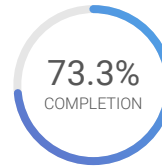
84 TH PLACE
OUT OF 3910
NATIONAL RANK

190 POINTS
OUT OF 310
PERFORMANCE SCORE

Average: 95.0 Points



Average: 59.5%



Average: 36.5%

Vulnerability Assessment
Analyst
Target Network Analyst
Cyber Operations Planner
Target Developer
Security Control Assessor

98th National
Percentile

Treasure Hunt (Easy)

90 POINTS
OUT OF 90

100.0%
ACCURACY

COMPLETION: **100.0%**

Perform a directory scan and identify hidden files on a remote HTTP server

DNS (Medium)

-10 POINTS
OUT OF 110

0.0%
ACCURACY

COMPLETION: **0.0%**

Perform a targeted service scan of a DNS server to identify the domain names that are blocked

Database (Hard)

110 POINTS
OUT OF 110

66.7%
ACCURACY

COMPLETION: **100.0%**

Perform a scan of a neo4j graph database server to identify the records stored on the database

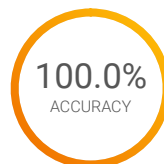
Web Application Exploitation Module

Identify actionable exploits and vulnerabilities and use them to bypass the security measures in online services.

89 TH PLACE
OUT OF 3910
NATIONAL RANK

120 POINTS
OUT OF 320
PERFORMANCE SCORE

Average: 104.0 Points



Average: 55.6%



Average: 35.8%

Cyber Operator
Software Developer
Exploitation Analyst
Systems Security Analyst
Database Administrator

98th National
Percentile

Hyperdash (Easy)

100 POINTS
OUT OF 100

100.0%
ACCURACY

COMPLETION: **100.0%**

Analyze source code on a web page and exploit local authentication measures

Tom's Login (Medium)

0 POINTS
OUT OF 100

0.0%
ACCURACY

COMPLETION: **0.0%**

Perform online password cracking by exploiting noSQL database querying vulnerabilities

Shipping (Hard)

20 POINTS
OUT OF 120

100.0%
ACCURACY

COMPLETION: **50.0%**

Exploit the web server to exploit an object deserialization vulnerability in order to achieve arbitrary remote code execution

